

Question 1: What is a conditional statement in programming?

A conditional statement is a structural control configuration that lets a program fork its execution path dynamically based on whether a specified boolean expression evaluates to true or false.

Question 2: What does an if statement do?

An `if` statement tests a boolean expression; if the expression evaluates to true, the program executes the block of code wrapped within its scope. If false, the block is completely skipped.

Question 3: What is the purpose of an else block?

It provides an alternative code execution route that runs exclusively if all preceding conditional tests within that evaluation chain resolve to a false state.

Question 4: When is an else block executed?

It triggers precisely when the associated parent `if` condition (and any matching `else if` conditions) fail and evaluate to false.

Question 5: What is a Boolean expression?

A relational or logical expression that evaluates strictly to one of two binary core states: `true` (represented by non-zero integers, typically 1) or `false` (represented by 0).

Question 6: Write the syntax of a simple if statement.

```
if (condition) {  
    // Execution statements go here  
}
```

Question 7: What operator is commonly used to test equality?

The equality verification operator token: `==`

Question 8: What does the modulus (%) operator return?

It calculates and returns the integer remainder of a division operation between two integer values.

Question 9: How can you check if a number is even?

By assessing whether the remainder of a division by 2 is exactly 0: `if (num % 2 == 0)`

Question 10: How can you check if a number is odd?

By validating whether the remainder of a division by 2 is not equal to 0: `if (num % 2 != 0)`

Question 11: What is the purpose of else-if?

It allows programs to sequence and evaluate multiple distinct conditions one after another after an initial `if` test fails, forming an exclusive branch network.

Question 12: What is a nested if statement?

An `if` statement declared inside the execution block of another existing conditional statement, used to test multiple layers of criteria.

Question 13: Can an if statement exist without an else?

Yes. An `if` block can operate on its own as an optional action. If the condition evaluates to false, execution passes directly to the next sequential line of code.

Question 14: What is the output when a false condition is tested in an if statement without else?

No code inside the condition's scope runs. The program skips the block entirely and moves down to the next instruction in the pipeline without executing or printing anything.

Question 15: What is the role of braces {} in C/C++ conditionals?

Braces group multiple individual instructions into a single composite execution block. If omitted, only the single line immediately following the condition is scoped under that conditional block.

Question 16: What does > mean?

The relational comparison operator tracking: **Greater Than**.

Question 17: What does < mean?

The relational comparison operator tracking: **Less Than**.

Question 18: What does >= mean?

The relational comparison operator tracking: **Greater Than or Equal To**.

Question 19: What does <= mean?

The relational comparison operator tracking: **Less Than or Equal To**.

Question 20: What does != mean?

The relational comparison operator tracking: **Not Equal To**.

Question 21: What is a switch statement?

A multi-way structural control selection component that routes execution path branches cleanly by matching an integer or character expression against list choice constants.

Question 22: What is a case label in a switch statement?

A matching value identifier prefix (`case literal:`) inside a switch block that targets and catches specific evaluated matches to run matching blocks of code.

Question 23: Why is break used in switch?

It exits the switch container block immediately. Without a break, execution falls through into subsequent case entries, running unrelated code blocks indiscriminately.

Question 24: What happens if break is omitted?

It triggers **fall-through behavior**. The application executes statements under subsequent cases sequentially until it encounters a break statement or reaches the end of the switch block.

Question 25: What is the default case in switch?

An optional fallback label that executes if none of the explicit case options match the evaluated switch expression. It acts like the final else block in a conditional ladder.

Question 26: Can switch compare strings in C++?

No. Standard C++ switch blocks match values against integral or enumeration structures exclusively. String literals cannot be matched directly inside a switch statement.

Question 27: Which statement is better for many constant choices: if-else or switch?

The `switch` layout is better because it is more readable and can be optimized by compilers into a jump table, outperforming long sequence chains of if-else checks.

Question 28: How do you test whether a value is positive?

By evaluating whether the parameter value is strictly greater than 0: `if (val > 0)`

Question 29: How do you test whether a value is negative?

By evaluating whether the parameter value is strictly less than 0: `if (val < 0)`

Question 30: How do you test whether a value is zero?

By checking equality against 0: `if (val == 0)`

Question 31: Write a condition to check if age is at least 18.

```
if (age >= 18)
```

Question 32: Write a condition to check if marks are above 50.

```
if (marks > 50)
```

Question 33: What is decision making in programming?

The design concept where developers configure software code logic to evaluate dynamic properties at runtime, changing execution routes to deliver contextual outcomes.

Question 34: Can nested if statements have multiple levels?

Yes. The nesting level limit is bounded only by compiler memory constraints. However, excessive nesting levels degrade code clarity and should be avoided when possible.

Question 35: What is logical AND (&&)?

A logical operator combining two boolean terms that evaluates to true if and only if both conditions match true, utilizing short-circuit optimization paths.

Question 36: What is logical OR (||)?

A logical operator combining two boolean terms that evaluates to true if at least one condition matches true, skipping subsequent tests if the first condition checks out true.

Question 37: What is logical NOT (!)?

A unary logic modifier operator that flips the conditional truth value of its target boolean clause expression.

Question 38: How do you check if a year is divisible by 4?

By evaluating whether the remainder of a division by 4 is exactly 0: `if (year % 4 == 0)`

Question 39: What is a leap year?

A calendar year containing 366 days instead of 365. This adjustment keeps calendar years aligned with the Earth's orbit, balancing orbital timing discrepancies over time.

Question 40: What is the first condition for a leap year?

The candidate year number must be evenly divisible by 4 (`year % 4 == 0`). However, centurial years must also pass secondary divisibility checks by 400 to qualify.

Question 41: How do you find the largest of two numbers?

Step	Details
1. Problem	Input two variables from a user session, compare their magnitudes, and display the larger value.
2. Steps	1. Start 2. Read parameters into val1 and val2 3. Test if val1 > val2; if true log val1, else log val2 4. Finish.
3. Dry Run	Provided details: val1 = 45, val2 = 67 Comparison check: 45 > 67 resolves to false → Routes to else path → Displays 67.
4. Source Code	<pre>#include <iostream> using namespace std; int main() { int val1, val2; cout << "Enter two distinct numbers: "; cin >> val1 >> val2; if (val1 > val2) { cout << "The largest value is: " << val1 << endl; } else { cout << "The largest value is: " << val2 << endl; } return 0; }</pre>
5. Expected Out	<pre>Enter two distinct numbers: 45 67 The largest value is: 67</pre>
6. Logic	Uses basic if-else relational expressions to branch execution flow based on which number holds the larger value.

Question 42: How do you find the largest of three numbers?

Step	Details
1. Problem	Capture three integers, evaluate their values, and determine the single largest numeric parameter.
2. Steps	1. Set parameters x, y, z 2. If $x \geq y$ and $x \geq z \rightarrow$ largest is x 3. Else if $y \geq x$ and $y \geq z \rightarrow$ largest is y 4. Else $\rightarrow$ largest is z.
3. Dry Run	Tracked inputs: x=14, y=52, z=33 Check 1: $14 \geq 52$ (false) Check 2: $52 \geq 14$ and $52 \geq 33$ (true) $\rightarrow$ Selects y $\rightarrow$ Outputs 52.
4. Source Code	<pre>#include <iostream> using namespace std; int main() { int x, y, z; cout << "Enter three distinct integers: "; cin >> x >> y >> z; if (x >= y && x >= z) { cout << "Largest: " << x << endl; } else if (y >= x && y >= z) { cout << "Largest: " << y << endl; } else { cout << "Largest: " << z << endl; } return 0; }</pre>
5. Expected Out	<pre>Enter three distinct integers: 14 52 33 Largest: 52</pre>
6. Logic	Combines relational markers using short-circuit logical AND (&&) operators to validate the largest value across multiple branches.

Question 43: What is an if-else-if ladder?

A chain of conditional statements used to test multiple sequential expressions in turn. The program steps down the ladder until a condition evaluates to true, executes that specific block, and exits the entire structure.

Question 44: What is the difference between if and switch?

- **if:** Evaluates range logic, inequality expressions, and complex multi-variable conditions combining logical symbols (`&&`, `||`).
- **switch:** Matches an expression against a list of constant integer or character values, generating optimized jump paths during compilation.

Question 45: Can switch use ranges directly?

No. Standard ISO C++ requires case expressions to match explicit constant values. Note that range extensions (e.g., `case 1 ... 5:`) are non-standard compiler-specific additions (like GCC) and break platform portability rules.

Question 46: What is fall-through in switch?

The structural behavior where code execution continues into subsequent case blocks because a `break` statement was omitted. This allows different case matches to share the same execution instructions.

Question 47: What data type is usually used for conditions?

The relational boolean condition testing type keyword: `bool` .

Question 48: Why are conditional statements important?

They remove linear limitations from software execution. Adding conditions lets programs adapt dynamically to real-time inputs, handle faults gracefully, and run complex business logic pathways.

Question 49: Give one real-life example of a conditional statement.

An automated ATM verification loop: **IF** a user's entered PIN matches the bank records, open the transaction menu; **ELSE** display an error notification and reject the card access block.

Question 50: Name four conditional structures covered.

1. Simple `if` conditional block structure
2. Standard paired `if-else` branching statement
3. Multi-branch sequential `if-else-if` ladder system
4. Constant value matching `switch-case` control component